

Homework 2

CHAPTER 8

1.2. Fill in the aggregate saving column in the following table. Use the data in the table to calculate the consumption function and the saving function, and plot these functions as well as the 45-degree line on a graph. What are the values for the MPC and the MPS ?

Aggregate Income, Y	Aggregate Consumption, C	Aggregate Saving, S
\$ 0	\$ 200	
100	250	
200	300	
300	350	
400	400	
500	450	
600	500	

1.4. Assume in a simple economy that the level of saving is -800 when aggregate output equals zero and that the marginal propensity to save is 0.25. Derive the saving function and the consumption function, and draw a graph showing these functions. At what level of aggregate output does the consumption curve cross the 45-degree line? Explain your answer and show that on the graph.

4.5. You are given the following data concerning Freedonia, a legendary country:

- (1) Consumption function: $C = 200 + 0.8Y$
- (2) Investment function: $I = 100$
- (3) $AE \equiv C + I$
- (4) $AE = Y$
 - (a) What is the marginal propensity to consume in Freedonia, and what is the marginal propensity to save?
 - (b) Graph equations (3) and (4) and solve for equilibrium income.
 - (c) Suppose equation (2) is changed to $I' = 110$. What is the new equilibrium level of income? By how much does the \$10 increase in planned investment change equilibrium income? What is the value of the multiplier?
 - (d) Calculate the saving function for Freedonia. Plot this saving function on a graph with equation (2). Explain why the equilibrium income in this graph must be the same as in part (b).

CHAPTER 9

1.3. Assume that in 2019, the following prevails in the Republic of Nurd:

$$Y = \$200$$

$$C = \$160$$

$$S = \$40$$

$$I \text{ (planned)} = \$30$$

$$G = \$0$$

$$T = \$0$$

Assume that households consume 80 percent of their income, they save 20 percent of their income, $MPC = 0.8$, and $MPS = 0.2$. That is, $C = 0.8Y_d$ and $S = 0.2Y_d$.

- Is the economy of Nurd in equilibrium? What is Nurd's equilibrium level of income? What is likely to happen in the coming months if the government takes no action?
- If \$200 is the "full-employment" level of Y , What fiscal policy might the government follow if its goal is full employment?
- If the full-employment level of Y is \$250, what fiscal policy might the government follow?
- Suppose $Y = \$200$, $C = \$160$, $S = \$40$, and $I = \$40$. Is Nurd's economy in equilibrium?
- Starting with the situation in part d, suppose the government starts spending \$30 each year with no taxation and continues to spend \$30 every period. If I remains constant, what will happen to the equilibrium level of Nurd's domestic product (Y)? What will the new levels of C and S be?
- Starting with the situation in part d, suppose the government starts taxing the population \$30 each year without spending anything and continues to tax at that rate every period. If I remains constant, what will happen to the equilibrium level of Nurd's domestic product (Y)? What will be the new levels of C and S ? How does your answer to part f differ from your answer to part e? why?

2.2. Suppose that the government of Ansonia is experiencing a large budget deficit with fixed government expenditures of $G = 250$ and fixed taxes of $T = 150$. Assume that consumers of Ansonia behave as described in the following consumption function:

$$C = 300 + 0.8(Y - T)$$

Suppose further that investment spending is fixed at 200. Calculate the equilibrium level of GDP in Ansonia. Solve for equilibrium levels of Y , C , and S . Next, assume that the Republican Congress in Ansonia succeeds in reducing taxes by 30 to a new fixed level of 120. Recalculate the equilibrium level of GDP using the tax multiplier. Solve for equilibrium levels of Y , C , and S after the tax cut and check to ensure that the multiplier worked. What arguments are likely to be used in support of such a tax cut? What arguments might be used to oppose such a tax cut?

CHAPTER 9 APPENDIX

1A.1. Assume the following for the economy of a country:

- Consumption function: $C = 60 + 0.75Y_d$
- Investment: $I = 75$
- Government spending: $G = 45$
- Net taxes: $T = -25 + 0.2Y$
- Disposable income: $Y_d \equiv Y - T$
- Equilibrium: $Y = C + I + G$

Solve for equilibrium income. How much does the government collect in net taxes when the economy is in equilibrium? What is the government's budget deficit or surplus?

CHAPTER 10

2.3. You are given this account for a bank:

Assets		Liabilities	
Reserves	\$1,200	Deposits	\$8,000
Loans	6,800		

The required reserve ratio is 10 percent.

- How much is the bank required to hold as reserves given its deposits of \$8,000?
- How much are its excess reserves?
- By how much can the bank increase its loans?
- Suppose a depositor comes to the bank and withdraws \$500 in cash. Show the bank's new balance sheet, assuming the bank obtains the cash by drawing down its reserves. Does the bank now hold excess reserves? It is meeting the required reserve ratio? If no, what can it do?

CHAPTER 11

2.4. Illustrate each of the following situations with a graph showing IS curve and the Fed rule, and explain what happens to the equilibrium values of the interest rate and output:

- A decrease in G with the money supply held constant by the Fed
- A decrease in G with the Fed changing M^s by enough to keep interest rates constant
- A decrease in P with no change in government spending
- An increase in Z with no change in government spending
- An increase in P and a decrease in G

3.1. Illustrate each of the following situations with a graph showing AS and AD curves, and explain what happens to the equilibrium values of the price level and aggregate output:

- A decrease in G with the money supply held constant by the Fed
- A decrease in the price of oil with no changes in government spending
- An increase in Z with no change in government spending
- An increase in the price of oil and a decrease in G